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SEMESTER – 6

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INFORMATION SECURITY

Unit -1 Introduction to Information Security

- **Introducton to Information Security**
 - What is Security
 - Types of Security
 - Key Information Security Components
 - Critical Characteristics of Information
 - Components of Information System
 - Balancing Information Security and Access
 - Security Professionals and the organization
 - The Need for Security
 - Business Needs First



When data is processed, organized, structured or presented in a given context so as to make it useful, it is called information.



[si-tyur-ə-tel]

A fungible, negotiable financial instrument that holds some type of monetary value.

UNIT -1 Inroduction to IS

- **What is “INFORMATION”?**

- Information is **organized or classified data**, which has some meaningful values for the receiver.
- Information is the **processed data** on which decisions and actions are based.

- **What is “SECURITY”?**

- “The **Quality or state of being secure** – to be free from danger”
- “**Protection against adversaries** – from those who would do harm, intentionally or other wise”

What is Information Security?



What is Information Security?

Information Security is not only about securing information from unauthorized access. Information Security is basically the practice of preventing unauthorized access, use, disclosure, disruption, modification, and destruction of information.

What is Information Security (InfoSec)?

A set of practices intended to keep data secure from unauthorized access or alterations



UNIT -1 Types of Information Security

- A successful organization should have the following physical layers of security in place to protect its operations.
 - **Physical Security**
 - **Personnel Security**
 - **Operations Security**
 - **Communications Security**
 - **Network Security**
 - **Information Security**

UNIT -1 Types of Information Security

- A successful organization should have the following physical layers of security in place to protect its operations.
 - **Physical Security**
 - To **protect physical items, objects or areas** from unauthorized access and misuse.
 - Building, Computers, Files etc....
 - **Personnel Security**
 - To **protect the individual or group of individuals** who are authorized to access the organization and its operations.
 - Admin, User etc....

UNIT -1 Types of Information Security

- A successful organization should have the following physical layers of security in place to protect its operations.
 - **Operations Security**
 - To protect the details of a particular operations or series of activities.
 - Product Manufacturing secret.
 - **Communications Security**
 - To protect communications media, technology and content
 - Wired / Wireless media, storage media etc.

UNIT -1 Types of Information Security

- A successful organization should have the following physical layers of security in place to protect its operations.
 - **Network Security**
 - To protect networking components, connections and contents.
 - Network Devices and Packets of data etc...

UNIT -1 Types of Information Security

- Information Security

Understanding the CIA Triad

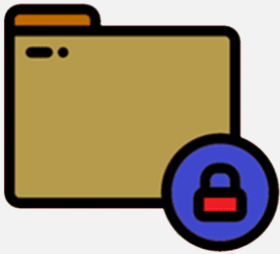


The CIA triad is a security model that has been developed in order to allow security professionals to think about and develop information security. ▲

UNIT -1 Types of Information Security

- A successful organization should have the following physical layers of security in place to protect its operations.
 - **Information Security**
 - To protect the **(CIA / AIC)**
 - **Confidentiality**
 - **Integrity and**
 - **Availability**of information assets.
 - The assets whether in **storage, processing or transmission .**

Confidentiality



- Cracking Encrypted Data
- Man In The Middle attacks on plain text
- Data leakage/
Unauthorised copying of sensitive data
- Installing
Spyware/Malware on a server

Integrity



- Web Penetration for malware insertion
- Maliciously accessing servers and forging records
- Unauthorised Database scans
- Remotely controlling zombie systems

Availability



- DOS/DDoS attacks
- Ransomware attacks –
Forced encryption of Key data
- Deliberately disrupting a server rooms power supply
- Flooding a server with too many requests

UNIT -1 Types of Information Security

- **Confidentiality**

- Confidentiality is roughly **equivalent to privacy**. Measures undertaken to ensure confidentiality are designed to **prevent sensitive information from reaching the wrong people, while making sure that the right people can in fact get it.**

Confidentiality

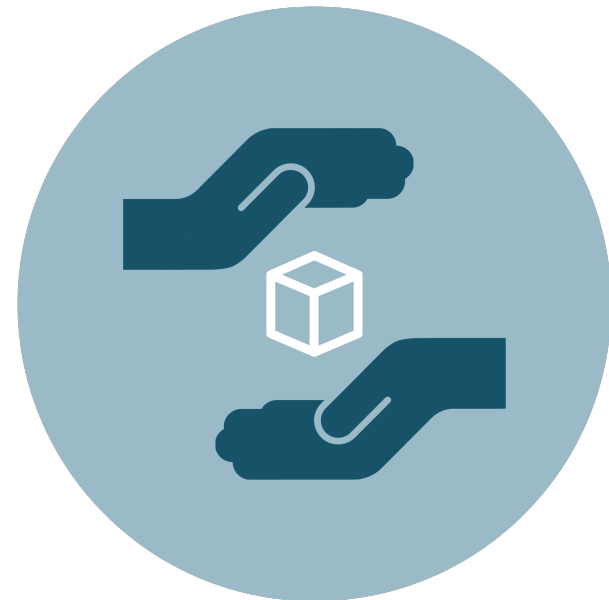


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- **Integrity**

- Integrity involves **maintaining the consistency, accuracy, and trustworthiness of data over its entire life cycle.** Data must not be changed in transit, and steps must be taken to ensure that data cannot be altered by unauthorized people

Integrity



UNIT -1 Types of Information Security

- **Availability**

- Availability of information refers to ensuring that **authorized parties are able to access the information when needed.**
- Information only has **value** if **the right people can access it at the right times.**

Availability



UNIT -1 Key Information Security Concepts

- Some Terms and Concept which will assist through IS
 - Access
 - A subject or object's **ability to use, manipulate, modify or affect** another subject or object.
 - Asset
 - The **Organizational resource** such
 - Logical (Website, Information, Data etc..)
 - Physical (Person, Computer etc..)

UNIT -1 Key Information Security Concepts

- Some Terms and Concept which will assist through IS
 - Attack
 - An intentional or unintentional **act that can cause damage to or otherwise compromise information** and/or the system that support it.
 - Active Attack-modification data
 - Passive Attack- watch, no change
 - Intentional Attack
 - Unintentional Attack
 - Direct Attack- IP identify
 - Indirect Attack- IP faked

UNIT -1 Key Information Security Concepts

- Some Terms and Concept which will assist through IS
 - Control, Safeguard or Countermeasure
 - **Security mechanisms, policies or procedures that can successfully counter attacks, reduce risk, resolve vulnerabilities and improve the security within an organization.**
 - Exploit
 - **A technique used to compromise a system. A threat agent may attempt to exploit a system or other information asset by using it illegally for their personal gain.**

UNIT -1 Key Information Security Concepts

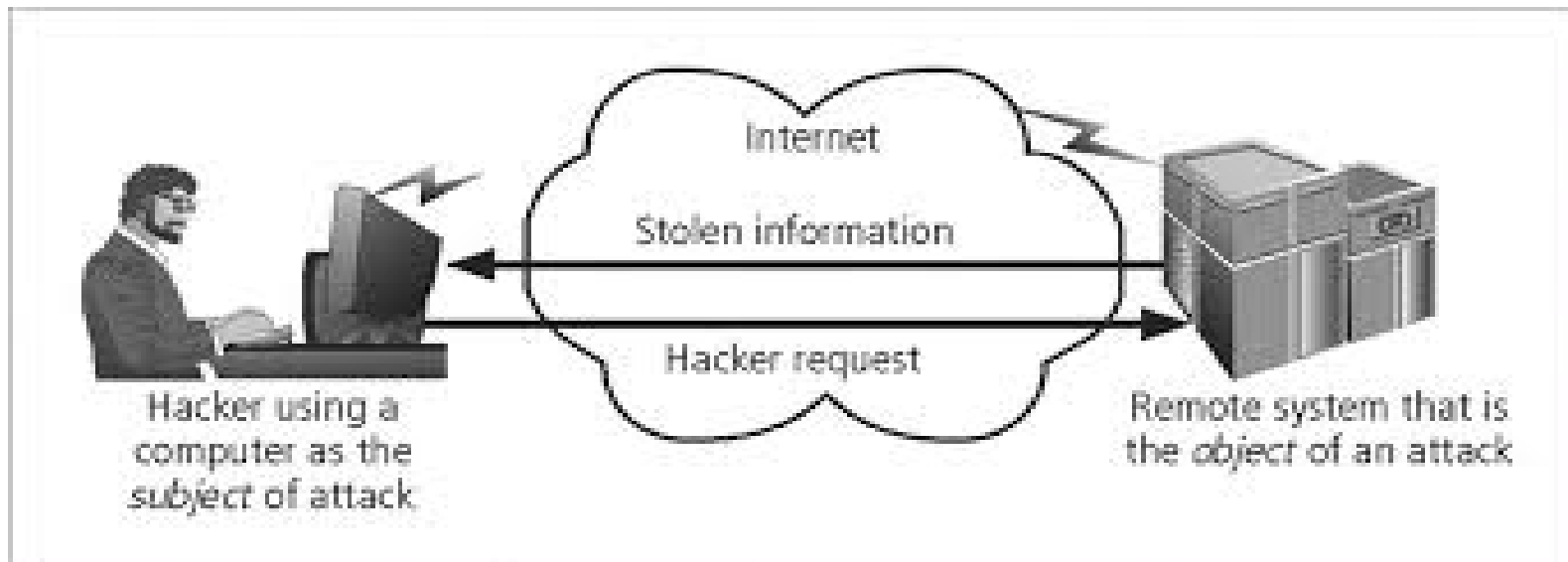
- Some Terms and Concept which will assist through IS
 - Exposure
 - A condition or state of being exposed.
 - Loss
 - **A single instance of an information asset suffering damage** or unintended or unauthorized modification or disclosure.
 - Protection Profile or Security posture
 - **The entire set of controls and safeguards, including policy, education, training and awareness and technology that the organization implements to protect the asset.**

UNIT -1 Key Information Security Concepts

- Some Terms and Concept which will assist through IS
 - Risk
 - **The probability that something unwanted will happen.**
 - Threat
 - **A category of objects, persons or other entities that presents a danger to an asset.**
 - Threat Agent
 - The specific instance or a component of a threat.

UNIT -1 Key Information Security Concepts

- Some Terms and Concept which will assist through IS
 - Subject and Objects
 - A computer can be either the subject of an attack or the object of an attack.



UNIT -1 Key Information Security Concepts

- Some Terms and Concept which will assist through IS
 - Vulnerability
 - **A Weakness or fault in a system or protection mechanism** that opens it to attack or damage.

UNIT -1 Critical Characteristics of Information

- The value of information comes from the characteristics it possesses.
 - Availability
 - Availability enables **authorized users – persons or computer systems – to access information without interference or obstruction** and to receive it in the **required format**.
 - Accuracy
 - Information has **accuracy when it is free from mistakes or errors** and it has the value that the end user expects.

UNIT -1 Critical Characteristics of Information

- The value of information comes from the characteristics it possesses.
 - Authenticity
 - Authenticity of information is the **quality or state of being genuine or original**, rather than a reproduction or fabrication.
 - Information is authentic when **it is the same state in which it was created, placed, stored or transferred**.

UNIT -1 Critical Characteristics of Information

- The value of information comes from the characteristics it possesses.
 - Confidentiality
 - Information has confidentiality when **it is protected from disclosure or exposure to unauthorized individuals or systems.**
 - It insured that **only those with the rights and privileges to access information** are able to do so.

UNIT -1 Critical Characteristics of Information

- The value of information comes from the characteristics it possesses.
 - Confidentiality
 - To **protect the confidentiality of** information, we can use number of measures like:
 - Information classification
 - Secure document storage
 - Application of general security policies
 - Education of information custodians and end users.

UNIT -1 Critical Characteristics of Information

- The value of information comes from the characteristics it possesses.
 - Integrity
 - Information has integrity when it is **whole, complete and uncorrupted.**
 - Utility
 - The **utility of information is the quality or state of having value** for some purpose. **Information has value when it can serve a purpose.**

UNIT -1 Critical Characteristics of Information

- The value of information comes from the characteristics it possesses.
 - Possession
 - **The possession of information is the quality or state of ownership or control.**
 - **Information is said to be one's possession if one obtains it, independent of format or other characteristics.**

UNIT -1 Components of an IS

- IS is much more than computer hardware. It is the **entire set of software, hardware, data, people, procedures and networks** that make possible the use of information resources in the organization.
- Software
- Hardware
- Data
- People
- Procedures
- Networks

UNIT -1 Components of an IS

- Software
 - The software component of the **IS comprises applications, operating systems and assorted command utilities**
 - Software is perhaps the **most difficult IS components to secure.**
- Hardware
 - Hardware is the **physical technology that houses and executes the software, stores and transports the data and provides interfaces for the entry and removal of information from the system.**

UNIT -1 Components of an IS

- Hardware
 - Physical security policies deal with hardware as a physical asset and with the **protection of physical assets form harm of theft.**
 - **Breach of Physical security can result in a loss of information.**

UNIT -1 Components of an IS

- Data
 - Data stored, processed and transmitted by a **computer system must be protected.**
 - Data is often the **most valuable asset possessed by an organization** and it is the main target of intentional attacks.
- People
 - Though often overlooked in computer security considerations, **people have always been a threat to information security.**

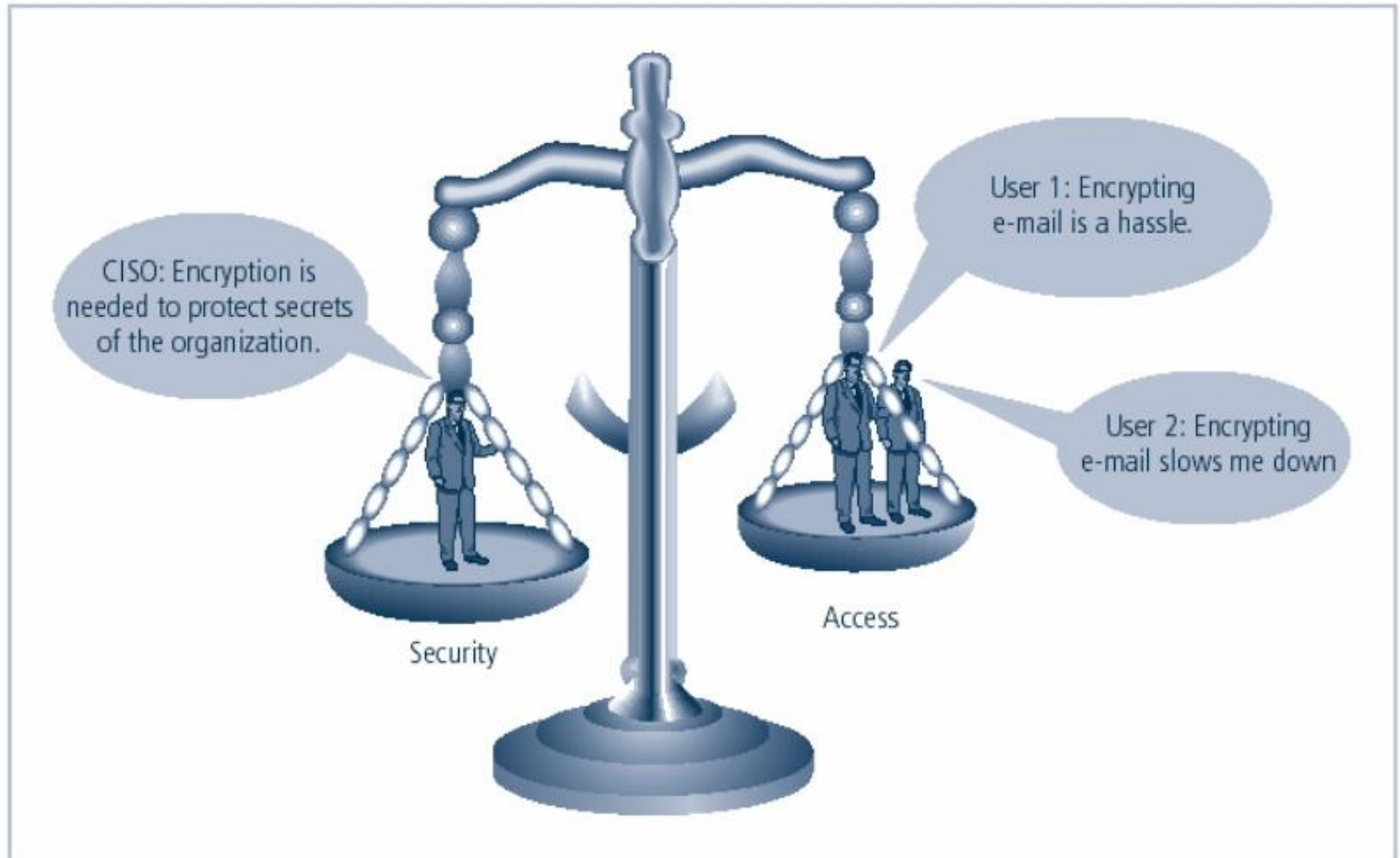
UNIT -1 Components of an IS

- Procedure
 - Procedures are written instructions for accomplishing a specific task.
 - When an unauthorized user obtains an organization's procedures, this poses a threat to the integrity of the information.
- Networks
 - The IS component that created much of the need for increased computer and information security is networking.

Balancing Information Security and Access:

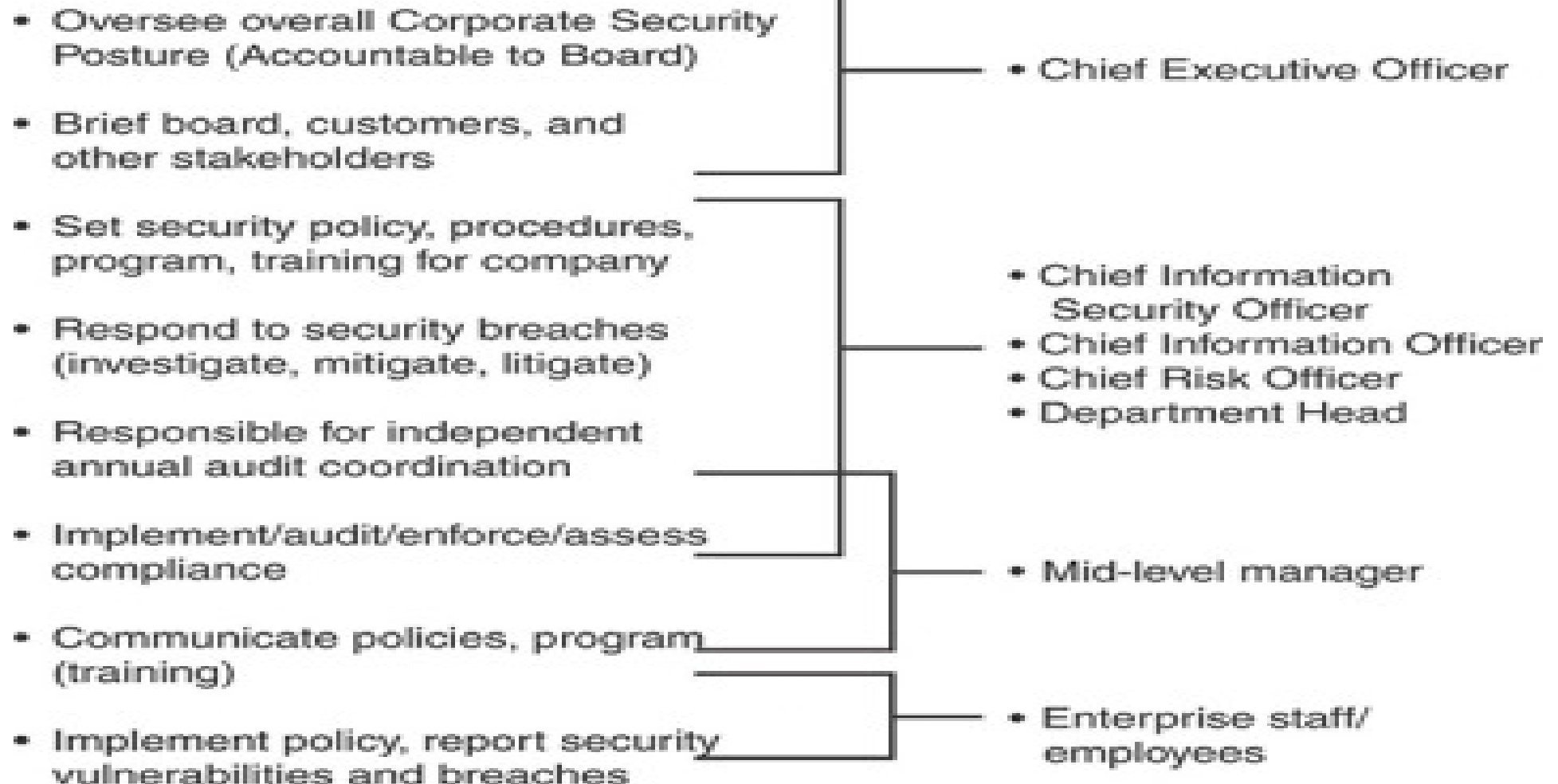
- Security is the sole purpose of the organisation to protect the interests of the users and to provide them with appropriate amount of information whenever necessary. Also, at the same time, it is necessary to provide adequate security to the information so that not anyone can access it. The need for maintaining the perfect balance of information security and accessibility arises from the fact that information security can never be absolute.
- It would be harmful to provide free access to a piece of information and it would be hard to restrict any accessibility. So, one needs to make sure that the exact required balance is maintained so that both the users and the security professionals are happy.

Balancing Information Security and Access



Responsibilities

Functional Role Examples



Security Professionals and the Organization

- ◆ Chief Information Officer
 - Senior technology officer
 - Primarily responsible for advising senior executive(s) for strategic planning
- ◆ Chief Information Security Officer
 - Individual primarily responsible for assessment, management, and implementation of securing information in the organization
 - May also be referred to as Manager for Security, Security Administrator, or a similar title

Security Professionals and the Organization (continued)

Information security project team should consist of individuals experienced in one or more facets of vast array of technical and nontechnical areas:

- ◆ Champion
- ◆ Team leader
- ◆ Security policy developers
- ◆ Risk assessment specialists
- ◆ Security professionals
- ◆ System, network, and storage administrators
- ◆ End users

UNIT -1 Business Needs (Why Business Needs IS?)

- Information Security performs four important functions for an organization.
 - (1)Protecting the **organization's ability to function.**
 - (2)Enabling the **safe operation of applications** running on the organization's IT systems.
 - (3)Protecting the **data the organization collects** and uses.
 - (4)Safeguarding the **organization's technology** assets.

UNIT -1 Business Needs

- Information Security performs four important functions for an organization.
 - (1) Protecting the **organization's ability to function**.
 - Both the **general management and IT management are responsible for implementing information security** that protects the organizations' ability to function.

UNIT -1 Business Needs

- Information Security performs four important functions for an organization.
 - (2) Enabling the **safe operation of applications** running on the organization's IT systems.
 - Organizations are **under immense pressure to acquire and operate integrated, efficient and capable application.**
 - A Modern organization **needs to create an envirement that safeguards these applications** – like Operating Syste, E-mails, Instant Messaging (IM) applications.

UNIT -1 Business Needs

- Information Security performs four important functions for an organization.
 - (3) Protecting the **data the organization collectes** and uses.
 - Any organization operating within the modern context of connected and **responsive services relies on information system.**
 - Therefore, protecting ***data information*** and ***data at rest*** are both critical aspects of IS.

UNIT -1 Business Needs

- Information Security performs four important functions for an organization.
 - (4) Safeguarding the **organization's technology assets**.
 - As an organization's **network grows to accommodate changing needs, more robust technology solutions** should replace security programs the organization has outgrown.
 - **An example of robust solution is a firewall....**

UNIT 1 COMPLETED